13 (a) Complete the table of values for $y = 2x^2 + 4x + 1$

$y = 2x^2 + 4x + 1$

For $x = -1$: $2(-1)^2 + 4(-1) + 1 = -1$

x	–3	–2	–1	0	1	2
y	7	1	-1	1	7	17

2 marks total
Square negative x before adding the other terms.

(2)

(b) On the grid opposite, draw the graph of $y = 2x^2 + 4x + 1$ for values of x from –3 to 2

2 marks total
Plot all six values on the facing grid; join with a smooth curve.

(2)

(c) Use your graph to find estimates for the solutions of $2x^2 + 4x - 3 = 1$

$2x^2 + 4x - 3 = 1$ means $2x^2 + 4x + 1 = 5$

2 marks total
Draw y = 5; read the two intersections.

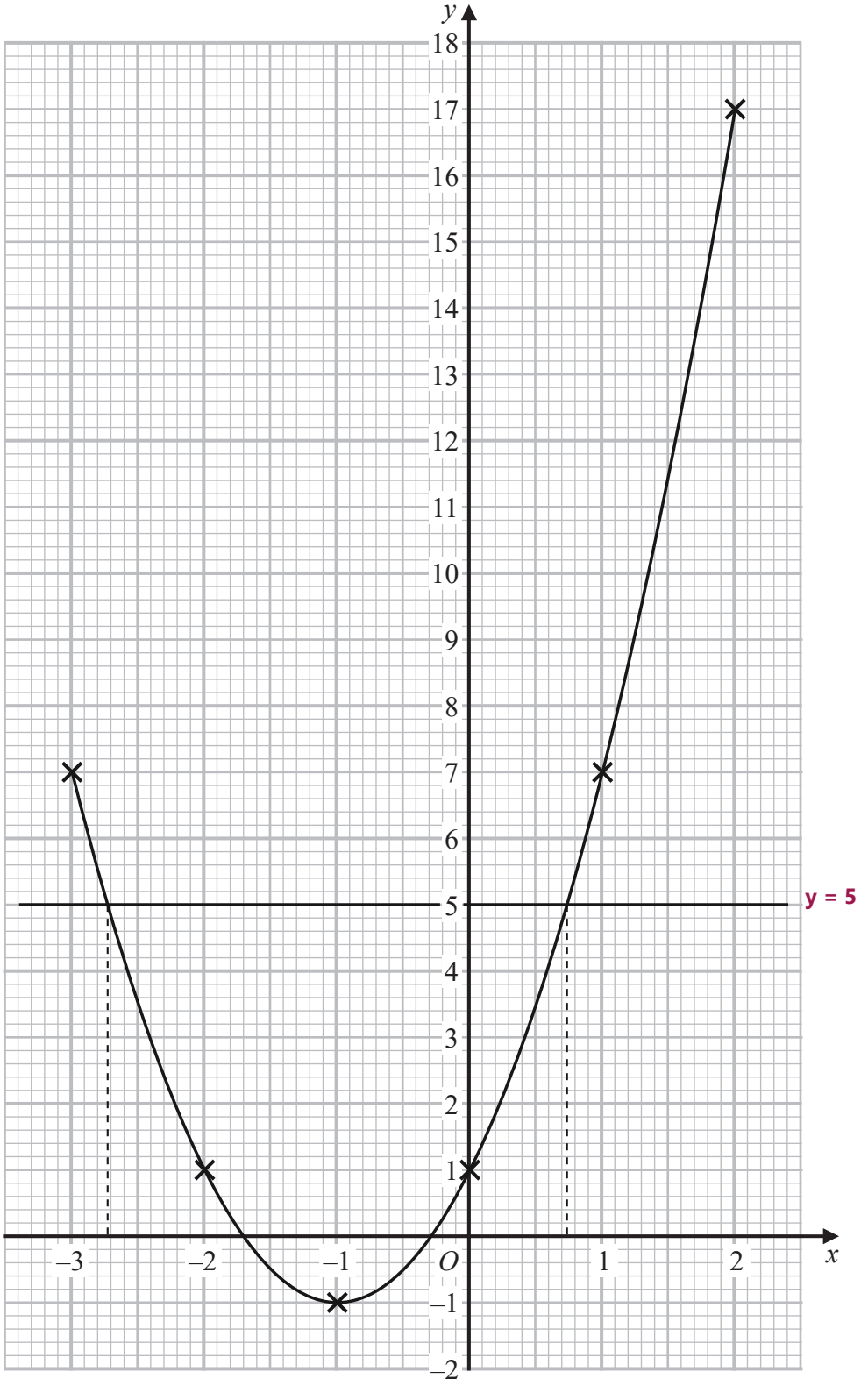
-2.7, 0.7

(2)

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(Total for Question 13 is 6 marks)